

Program : Diploma in Biomedical Engineering	
Course Code : 5243A	Course Title: Analytical Instruments
Semester : 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4, T:0, P:0)	Periods per semester: 60

Course Objectives:

- To impart knowledge on the fundamentals of Analytical instruments
- To equip students to handle various analytical instruments which are useful for clinical analysis in hospitals and pharmaceutical laboratories.

Course Prerequisites: Nil

Course Outcomes:

On completion of the course, the student will be able to:

CO_n	Description	Duration (Hours)	Cognitive Level
CO1	Describe the fundamentals of Analytical Instrument.	14	Understanding
CO2	Explain UV, Visible, IR Spectrophotometers and flame photometer	16	Understanding
CO3	Illustrate the instruments for Chromatography, Electrophoresis and densitometry.	14	Understanding
CO4	Explain pH meter, blood cell counters and blood Gas Analyzer	14	Understanding
	Series Test	2	

CO – PO mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Describe the fundamentals of Analytical Instruments.		
M1.01	Describe the elements of Analytical Instrument and its performance requirement	7	Understanding
M1.02	Summarize the Instrument Calibration Techniques & Validation	3	Understanding
M1.03	Explain Beer-Lambert's law and colorimeter	4	Understanding
Contents: Fundamentals of Analytical Instruments Introduction to Analytical Instrumentation: Fundamentals of analytical instruments: Elements of an analytical instrument - PC based analytical instruments - Classification of instrumental techniques. Performance requirement-definition of accuracy, precision, sensitivity, resolution and error, need for calibration Analytical instruments used in clinical environment (listing) and its applications. Electromagnetic radiation - Electromagnetic spectrum- Laws relating to absorption of radiation. Beer-Lambert's Law. Colorimeters - block diagram, Principle & applications, Calibration procedure.			
CO2	Explain UV, Visible, IR Spectrophotometers and flame photometer		
M1.01	Explain basic components of spectrophotometer	5	Understanding
M2.01	Describe the working principle of UV, Visible and IR spectrophotometers	9	Understanding
M2.02	Illustrate the working principle and constructional details of flame photometer.	2	Understanding
	Series Test – I	1	
Contents: SPECTROPHOTOMETER Principle of absorption spectroscopy- Types(Listing) Spectrophotometers: basic sub components - monochromator, detection systems and amplifiers - single beam & double beam - visible, UV & IR (basics only), - applications in biochemical analysis. Flame Photometry: Principle and constructional details of flame photometer- Emission system - Optical system - Detectors. Applications of flame photometry.			

CO3	Illustrate the instruments for Chromatography, Electrophoresis and densitometry.		
M3.01	Explain chromatography and its types.	3	Understanding
M3.02	Explain the principle and construction details of gas and liquid chromatography in detail.	6	Understanding
M3.03	Describe the working of electrophoresis machine and densitometer	5	Understanding
Contents: Chromatography And Electrophoresis Chromatographic process - Classification- Terms in chromatography - Gas chromatography: Block diagram - Principle - Constructional details Liquid Chromatography: Types of liquid chromatography- High pressure Liquid Chromatography (HPLC): Principle - Constructional details, Applications in medical field. Electrophoresis apparatus - basic principle and block diagram. Densitometer - principle and block diagram			
CO4	Explain pH meter, blood cell counters and blood Gas Analyzer		
M4.01	Explain the principle & construction of pH meter.	3	Understanding
M4.02	Detail the working of blood cell counters and Hb meter.	6	Understanding
M4.03	Describe the principle & construction of Blood Gas Analyzer and glucometer	5	Understanding
	Series Test – II	1	
Contents: pH METER, BLOOD CELL COUNTERS & BLOOD GAS ANALYZER pH meter - Principle of pH measurement - Electrodes for pH measurement - Block diagram of pH meter. Types of blood cells, Blood cell counters - types - coulter counter and flow cytometry. Auto analyzers - block diagram and working. Hb meter - block diagram Blood Gas Analyzer - pO ₂ and pCO ₂ measurement - Principle, construction and working of PO ₂ electrode and PCO ₂ electrode - Blood Gas Analyzer - block diagram of blood gas analyser. Glucometer - working principle and basic block diagram			

Text / Reference:

T/R	Book Title/Author
T1	R.S. Khandpur, 'Handbook of Analytical Instruments', Tata McGraw Hill publishing Co. Ltd., 2003.
T2	G.W. Ewing, Instrumental Methods of Analysis, Mc Graw Hill, 2004.
R1	Robrt D Braun', 'Introduction to Instrumental Analysis' , Mc Graw Hill Ltd
R2	Galen W Ewing, 'Instrumental methods of chemical analysis' , Mc Graw Hill Ltd

Online Resources:

Sl.No	Website Link
1	https://www.hemakumar.in/analytical-instrumentation
2	https://www.slideshare.net/ShreyaAhuja2/ph-meter-54084084
3	https://www.ktuassist.in/2020/06/ktu-ae402-analytical-instrumentation.html#